SEI KANG



Research Area: Virtual/Augmented Reality, Virtual Agent, HCI

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EDUCATION

Sep 2025 – Current	**Ph.D. in Department of AI Convergence, Chonnam National University, Republic of Korea** [Affiliation Lab] Empathic Computing Lab in Korea (Advisor: Seungwon Kim)
Mar 2024 – Aug 2025	**M.S. in Department of AI Convergence, Chonnam National University, Republic of Korea** [Affiliation Lab] Empathic Computing Lab in Korea (Advisor: Seungwon Kim)
Mar 2020 – Feb 2024	**B.S. in** Department **of Software Engineering,** Chonnam National University, Republic of Korea [Affiliation Lab] Empathic Computing Lab in Korea (Advisor: Seungwon Kim) GPA: 4.42/4.5

PUBLICATION (REGULAR PUBLICATIONS)

Exploring the Effects of Support Type and Task Difficulty for Virtual Assistants as Social Companions
Sei Kang, YunSu Lee, Gun A. Lee, Soo-Hyung Kim, Hyung-Jeong Yang, Ji-eun Shin, and Seungwon Kim
Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI)

Effects of Gesture Size in Virtual Agents on Persuasive Communication
Gayun Suh, Gun A. Lee, Hyung-Jeong Yang, Soo-Hyung Kim, Ji-eun Shin, Jaejoon Jeong, **Sei Kang**, and Seungwon Kim
Proceedings of the 31th ACM Symposium on Virtual Reality Software and Technology (VRST)

Social resources facilitate pulling actions toward novel social agents than pushing actions in virtual reality

Jaejoon Jeong, Hwaryung Lee, Ji-eun Shin, Daeun Kim, **Sei Kang**, Gun A. Lee, Soo-Hyung Kim, Hyung-Jeong Yang, Seungwon Kim
Empathic Computing 2 (1), 202514-202514

Design and Evaluation of a Virtual Agent for Interpersonal Emotion Regulation in VR
Sei Kang, Gun A. Lee, Hyung-Jeong Yang, Soo-Hyung Kim, Ji-eun Shin, Jaejoon Jeong, Myungho Lee, and Seungwon Kim
2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

Three Techniques for Enhancing Emotional Expression on Embodied Avatar Face in VR
Jaejoon Jeong, Gun A. Lee, Hyung-Jeong Yang, Soo-Hyung Kim, Ji-eun Shin, Gayun Suh, **Sei Kang**, and Seungwon Kim
2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

Performance Analysis of GazeRayHand Techniques for Complex Object-Manipulation Tasks in VR Environments
Yunsu Lee, **Sei Kang**, and Seungwon Kim
Journal of Digital Contents Society - KCI, 26(8), 2237-2247. (2025)

GazeRayHand: Combining Gaze Ray and Hand Interaction for Distant Object Manipulation
Sei Kang, Jaejoon Jeong, Soo-Hyung Kim, Hyung-Jeong Yang, Gun A Lee, and Seungwon Kim
Applied Sciences - 15(13), 7065. (2025)

The RayHand Navigation: A Virtual Navigation Method with Relative Position between Hand and Gaze-Ray
Sei Kang, Jaejoon Jeong, Gun A. Lee, Soo-Hyung Kim, Hyung-Jeong Yang, and Seungwon Kim
Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI)

PUBLICATION (POSTER/ADJUNCT)

Willingness to Interact: Social Resources Facilitate Pulling Actions toward Social Avatars in Virtual Reality
Jaejoon Jeong, Hwaryung Lee, Ji-eun Shin, Daeun Kim, **Sei Kang**, Gun A. Lee, Soo-Hyung Kim, Hyung-Jeong Yang, and Seungwon Kim
In Proceedings of 2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)

HONORS AND AWARDS

2026.09 - 2028.08 **NRF Ph.D. Fellowship (RS-2026-25574236)**
National Research Foundation, Korea

2026 -	**Ph.D. Excellence Scholarship (Science and Engineering)** Korea Student Aid Foundation (KOSAF)
2026	**2025 BK21 Fellowship Scholarship** Chonnam National University
2025	**Best Presentation Award** APMAR 2025
2024.09 - 2025.08	**NRF M.S. Fellowship (RS-2024-00465826)** National Research Foundation, Korea
2024.03 - 2025.08	**Presidential Science Scholarship (M.S.)** Korea Student Aid Foundation (KOSAF)
2023	**Outstanding Paper Award** Korea Digital Content Society
	Outstanding Paper Award Korea Society for Smart Media
2022~2023	**National Excellence Scholarship (Science and Engineering)** Korea Student Aid Foundation (KOSAF)
2022	**Grand Prize, Web Portfolio - AI+X Coding Fundamental Competency Enhancement Competition** Gwangju Jeonnam Regional Innovation Platform

PATENTS

Method and System for Distant Object Manipulation using Gaze Ray and Hand Interface
Korea Granted Patent (Registration No. 10-2864482)

An Electronic Device and Its Operation Method That Provides a Steering-Based Virtual Reality (VR) Navigation Environment That Operates Through a RayHand Control Method That Combines a Gaze and a Hand
Korea Granted Patent (Registration No. 10-2783127)

INVITED TALKS

Oct 2025 — **The 17th Asia-Pacific Workshop on Mixed and Augmented Reality (APMAR) 2025**
Pitch Your Work

- Topic: Design and Evaluation of a Virtual Agent for Interpersonal Emotion Regulation in VR

Oct 2024 **IEEE International Symposium for Mixed and Augmented Reality (ISMAR) 2024**
Lightning Talk of the LocXR Workshop

- Topic: The RayHand Navigation: A Virtual Navigation Method with Relative Position between Hand and Gaze-Ray

Jan 2024 **Korea Computer Congress (KCC) 2024**
Oral Presentation at Top Conference Session

- Topic: The RayHand Navigation: A Virtual Navigation Method with Relative Position between Hand and Gaze-Ray

ACADEMIC SERVICE

2026 **Reviewer**
VRST 2026 poster, IEEE TVCG

2025 **CG&I Society, Metaverse+AI+XR Workshop (KSC 2025)**
Steering Committee Chair

2022 **Korea Computer Graphics Society (KCGS) 2022**
Student Volunteer

TEACHING EXPERIENCES

Fall 2026 **TA, CIS3003 Artificial Intelligence**
Chonnam National University

Spring 2026 **TA, SAI0011 Software Capstone Design**
Chonnam National University

Fall 2025 **TA, CIS3020 Computer Graphics**
Chonnam National University

Fall 2024 **TA, CIS3003 Artificial Intelligence**
Chonnam National University

Spring 2024 **TA, STT3019 Java Programming and Lab**
Chonnam National University

SKILLS

Programming	C, C++, C#, Python, Java, JavaScript
Development tools	**Unity (Proficient)**, Unreal